

Learning to Open and Remove: Evolving Robot Control Code for Laboratory Manipulation

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Abstract—Opening laboratory equipment and removing a container cap require more than reaching and closing a gripper. The robot must change an object’s constraint while distinguishing successful release from loss of contact. We study how a human-guided coding agent acquires these operations by revising executable perception, control, and verification code from physical feedback. Two case studies use a shared laboratory robot stack but different action supervision. For an incubator door, twelve successful teleoperation episodes selected from 57 candidates are compiled into a contact-relative trajectory with depth-based orientation alignment and grasp checks. For a culture-media bottle cap, no cap-specific teleoperation demonstration is supplied; one target click and seven calibration observations support a local visual controller. Recorded results show a verified open-door endpoint, a separate verified closing run, and cap removal followed by 107.4mm of held transport. Failed and successful door pulls both end with nearly closed, empty jaws, demonstrating why gripper state alone is an inadequate task evaluator. The contribution is an auditable account of how physical failures changed the representation and verification of manipulation, not a benchmark-wide success-rate claim. The resulting controllers execute without a language model in the servo loop.

I. INTRODUCTION

An incubator door and a bottle cap are small obstacles to an automated biological workflow, yet each exposes a gap between robot motion and useful physical work. A gripper can close without engaging a recessed handle. It can briefly retain contact and then slide off. A cap can appear to rise because the entire bottle moved. A command trace that looks correct is therefore not enough: the robot must verify what changed in the world.

We call the common structure *constraint-release manipulation*. The robot engages an object, moves it relative to a supporting body, and establishes a new access state. A door remains constrained by its hinge but leaves the closed configuration. A cap separates from the bottle opening and must remain held afterward. This definition groups the tasks by their required state transition; it does not assume a measured break-away force, a particular latch, or a threaded cap.

The central challenge is that evidence of grasping and evidence of task completion can disagree. In our door trials, both failed pulls and successful opening endpoints were followed by an almost fully closed gripper. The same signal meant that contact was absent, but did not reveal whether the door had opened. Conversely, a partially closed gripper did not prove that the cap had separated from a stationary bottle. These ambiguities motivated changes to the code that acquires observations, selects motion, and judges outcomes.

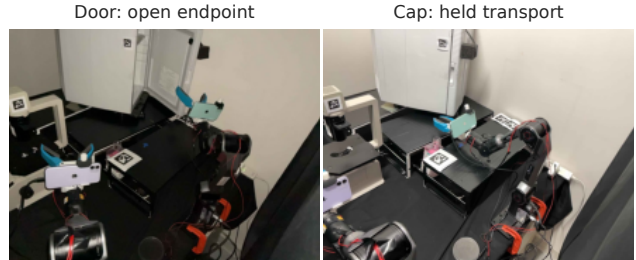


Fig. 1. Two laboratory access operations. Left: the door is open at the verified endpoint of T7. Right: the cap remains held after transport, with the bottle mouth exposed. Images show outcomes, not repeated trials.

Coding agents offer a way to revise these mechanisms together. Instead of only modifying action-network weights, they can inspect images, mine archived demonstrations, fit geometric models, write numerical controllers, and change when a trial is allowed to continue. We investigate this process in an existing laboratory cell, with human task instructions and intermittent corrections. The agent was not given a finished door-opening or cap-removal policy. It was given a working robot repository and access to observations and tools.

Our contribution has three parts. First, we formulate the distinction between *engagement*, *release*, and *retention* as explicit evidence requirements. Second, we describe two executable controllers assembled through physical feedback: demonstration-grounded door manipulation and cap removal without a task-specific action demonstration. Third, we reconstruct the failure-to-code-change chain using saved observations, source history, and runtime records. This is a retrospective two-case study, not a controlled comparison against alternative learning algorithms.

II. RELATED WORK

Laboratory automation. A mobile robotic chemist demonstrated how a robot can operate conventional laboratory equipment within an autonomous experimental search [1]. Our focus is earlier in that pipeline: acquiring and validating the small manipulation capabilities that make equipment and containers accessible. We optimize neither an assay nor a biological hypothesis in these experiments.

Code-generating robot agents. Code as Policies composes executable robot behavior from language-generated programs [2]. VoxPoser combines language-guided spatial reasoning with model-based trajectory generation and can incorporate contact experience [3]. ReAct exemplifies the

broader reasoning–action feedback pattern [4]. We focus on how physical failure changes a persistent controller and its evidence requirements, rather than on a single generated plan.

Physical policy improvement. ENPIRE constructs reset and verification interfaces through human-guided coding agents, then exposes fixed environment APIs for autonomous policy improvement [5]. It supports heuristic, imitation, reinforcement-learning, and hybrid approaches. Our study shares the physical feedback premise; it examines laboratory access operations and traces changes to observation and verification alongside control. We do not claim that tool use or agent-generated environment code is absent from ENPIRE.

Learned motor policies. OpenVLA and $\pi_{0.5}$ demonstrate the value of pretrained visuomotor policies and broad generalization [6], [7]. Our cap result uses zero *cap-specific* action demonstrations and no task-specific neural-policy training; it does not imply that VLAs require a new demonstration for every deployment. Program adaptation is complementary: a learned motor policy could be one component of the same execution stack.

Perception and simulation. SAM provides general-purpose segmentation [8], while MuJoCo supports model-based kinematics and contact analysis [9]. Both were available in the broader development environment. The two accepted runtime paths instead used classical image operations, registered depth, and mechanical state for their critical evidence. The relevant capability was selecting suitable tools, not maximizing their number.

III. TASK AND EVIDENCE MODEL

A. What must change?

Let o_t contain images, depth, and measured robot state. Let g_t denote engagement evidence, r_t a task-specific release state, and h_t continued retention. These are predicates over observations, not synonyms for issued commands. The cap goal requires $r_t \wedge h_t$: removal without dropping the cap. The door goal requires an independently observed open state; continuous handle retention is desirable but not necessary to label the final door state.

This distinction changes recovery. An empty grasp after a pull cannot safely trigger another identical pull without first inspecting the door. Similarly, opening the jaws at a destination does not establish stable cap placement. The program must stop the claim at the last supported stage.

B. Development and execution are separate loops

During development, Codex reads observations and code, proposes a revision, and tests a bounded physical segment. The operator supplies task intent and semantic or geometric corrections when needed. Successful behavior is retained as an executable artifact with configuration and tests. During execution, a deterministic state machine calls camera, geometry, and motion interfaces; it does not ask the language model to produce every servo command.

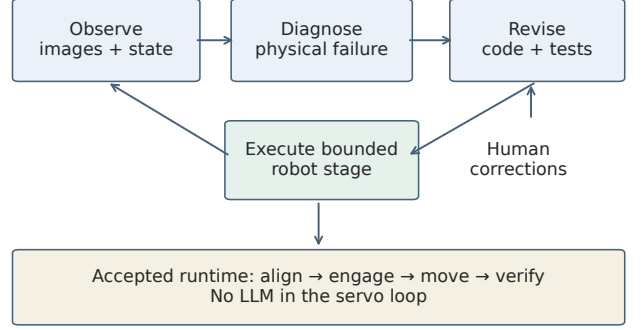


Fig. 2. Development changes executable code; deployment runs the resulting controller. Human corrections enter development. Saved physical observations remain separate from editable code and retrospective analysis.

TABLE I
ACTION SUPERVISION AND ADDITIONAL RESOURCES. A DEMONSTRATION IS AN EPISODE, NOT A SINGLE POSE.

Resource	Door opening	Cap removal
Task action demos	12 successes selected from 57 archived episodes	0 cap-specific episodes
Human input	Task instruction, archive location, corrections	One target click, semantic/geometric corrections
Physical data	Contact, proof, pull, endpoint observations	Seven calibration poses; approach, lift, transport observations
Shared tools	RGB-D/RGB, robot state	OpenCV, MuJoCo/Mink

We represent a development artifact as $C = (P, V, X)$, where P specifies target interpretation and motion, V specifies evidence checks, and X connects observation and execution services. A physical failure can require a change to any of these, not just a numeric gain. For example, replacing a shadow-based feature changes P ; adding a door-state check changes V ; isolating a fragile camera process changes X .

C. Resources and supervision

The system used a bimanual Piper installation with a position-controlled gripper, fixed-head Record3D RGB-D, and RGB wrist cameras. The reported manipulation used the right arm. The existing repository supplied robot RPC, CAN control, forward/inverse kinematics, gripper calibration, and camera interfaces. MuJoCo/Mink supplied model-based checks; fixed fiducials supported camera registration. The agent also had shell, Python, numerical libraries, tests, version history, and prior development context.

The operator explicitly introduced the archived door recordings before their use. The agent selected successful examples and compiled them; it did not request newly recorded demonstrations in this experiment. Door closing used a separate closing demonstration. Cap development used no cap-specific action trajectory but did use physical data and

human feedback. Thus neither case is data-free or human-free. Their commonality is the development mechanism, not a claim that the completed door controller was transferred to the cap.

IV. DEMONSTRATION-GROUNDED DOOR MANIPULATION

A. From recordings to a contact-relative pull

Twelve archived teleoperation episodes were accepted after inspection of terminal head images. The compiler chose an actual medoid episode using contact position and orientation, rather than averaging incompatible trajectories. Its demonstrated contact pose T_c^d defines a relative motion:

$$\Delta T_k = (T_c^d)^{-1} T_k^d, \quad T_k^{\text{run}} = T_c^{\text{run}} \Delta T_k. \quad (1)$$

The selected motion contained 148 samples from closure to release, spanning approximately 4.93 s in the recording. The live contact anchor therefore changes the trajectory’s placement without replacing its articulated shape. Demonstration-derived visual/pose pairs also supported a local correction model. Demonstrations supplied motion structure, not a complete success test.

Initially, the live starting pose was incorrectly treated as a contact reference. Saved observations exposed a large offset. The accepted approach instead used a demonstrated pre-close pose, a current door-plane orientation, and a bounded visual correction. Eight head RGB-D frames supported vertical plane fitting with RANSAC; the measured yaw was compared with a reference yaw. The right-camera red appliance label served only as an auxiliary rigid-body feature, not as the recessed handle itself. Corrections were limited to three steps of at most 15 mm each. A dark, claw-like feature was rejected after the operator identified it as a shadow.

B. Engagement before sustained pulling

At contact, the controller closed once and sampled measured gripper opening at 20 Hz. Opening is normalized as $a = 1$ for open and $a = 0$ for closed. It is not pressure. The settled closure had to exceed the empty-grasp bound 0.02 and remain stable over the final ten samples. A 5 mm proof pull then tested whether at least 65% of the closure opening remained, followed by a stationary reverification. A short proof reduced obvious empty pulls but was not assumed to guarantee long-horizon retention.

For the full pull, end-effector targets were sent through the established teleoperation command path at 30 Hz. The accepted implementation used twice the demonstration duration and checked opening every 15 frames. This was a position-target controller, not a break-away force estimator. If opening fell below the full-pull retention threshold, motion stopped, the arm retreated 15 mm, and the jaws opened. The controller then observed the door rather than blindly attempting another pull.

C. Verifying the object, not just the gripper

The endpoint evaluator registered head RGB-D to stored open and closed references using fixed tags. It compared depth evidence in a door-state region and allowed an unknown outcome when evidence was insufficient. Arm pixels contaminating that region motivated a subsequent mask correction. References and observations were kept distinct from the policy’s expectation of success.

A separate close demonstration was necessary. Reversing the opening path missed the effective push location by about 15 cm in height. The accepted closing routine used an open-jaw push from the dedicated close recording, then checked the closed endpoint. Opening and closing share an appliance but need not share a reversible contact policy.

V. CAP REMOVAL WITHOUT AN ACTION DEMONSTRATION

A. Grounding target, jaws, and supporting bottle

Global white-object detection confused the cap with gripper and laboratory components. One operator click anchored the intended instance. OpenCV color, connected-component, and geometric operations then represented a small white cap above the colored bottle neck. Jaw midpoint and jaw span provided a gripper-relative target geometry. Registered RGB-D localized the selected surface, while a completed bottle/cap model represented geometry not fully visible in depth.

The supporting bottle was recessed between platforms rather than standing on the assumed tabletop. This changed approach ordering: align laterally in free space, then descend, then close. MuJoCo/Mink checked candidate motion against the scene and robot geometry. Simulation was a critic, not ground truth: one model-only collision disagreement was later scoped to the exact route already supported by hardware evidence, rather than disabling collision checks for new paths.

B. Learning a local image-to-motion correction

Seven open-jaw physical observations supported a local correspondence between robot displacement and image error. With jaw-relative error e and Cartesian displacement Δx , the controller used a local linear model

$$\Delta e \simeq J \Delta x, \quad \Delta x = -J^+ e, \quad (2)$$

with bounded corrections and fresh observations. This calibration learned a local control relationship, not a general visual policy. The final waypoint sequence preserved the physically checked approach and egress.

Before lift, the target had to lie within the jaw segment and within 0.25 jaw spans perpendicular to it. Opening had to lie in the calibrated non-empty band $[0.10, 0.40]$. The colored bottle support could move by no more than 0.10 of its visual scale. These normalized image checks avoided defining success by a fixed pixel count, although the detector remained specific to the enrolled scene and appearance.

TABLE II

DOOR OPENING OBSERVATIONS. a_c : SETTLED CLOSURE; a_p : PROOF SUMMARY; a_e : LATER EMPTY-GRASP OBSERVATION. MISSING VALUES ARE NOT FILLED.

Trial	a_c	a_p	a_e	Endpoint evidence
T1	.412	.366	.003	Closed
T2	.407	.365	.003	Closed
T3	.366	.366	–	Not classified here
T4	.323	.319	.004	Not classified here
T5	.293	.291	–	Not classified here
T6	.407	.290	.003	Open observed
T7	.306	.296	.003	Open verified

C. Removal and retention require different witnesses

The removal check combined measured vertical rise, persistent opening, visible jaw motion, limited bottle displacement, and reduced cap-like appearance at the source. The transport check added measured displacement and aperture stability. A commanded release was not accepted as placement without evidence of a stable destination support. This separation prevented an unverified final step from invalidating, or being confused with, the verified earlier capability.

VI. RECORDED RESULTS AND FAILURE ANALYSIS

A. Protocol and unit of evidence

We retrospectively aligned source revisions, command records, and immutable observations. For the door, E1–E3 are early contact-only attempts and T1–T7 are physical pull attempts, not code versions. For the cap, the principal quantitative result is one before-lift/after-lift/after-transport capture triplet. The record does not contain a randomized repetition study. We therefore report observed endpoints and measured changes, not a success-rate estimate from these selected development records.

B. A non-empty proof did not predict the endpoint

The first structured door pull, T1, retained opening from 0.412 after close to 0.366 after the proof. Yet its endpoint did not support opening. T2 showed a similar pattern. Slower execution and checkpoints made subsequent failures detectable but did not by themselves establish success. Later yaw-aligned T6 and autonomous T7 reached open endpoints, although grasp was absent afterward (Table II). The exact instant at which opening and slip occurred relative to each other is not reconstructed from the sparse record.

T7 began from a verified closed state. Its live plane yaw was -4.61° , giving approximately 0.775° residual correction relative to its reference; one visual correction was approximately 3.5 mm. After proof and stationary recheck, the full pull stopped on an empty-grasp observation. The registered endpoint was open. A separate autonomous closing run subsequently verified the closed state. This establishes the recorded access-state changes, not continuous grasp throughout opening or reliable repeated cycling.

TABLE III

CAP STATE EVIDENCE FROM ONE RECORDED SEQUENCE. TRANSPORT IS MEASURED FROM THE POST-LIFT POSE; APERTURE IS AN OPENING RATIO, NOT FORCE.

State	EE height (mm)	Opening	Source ratio
Before lift	734.35	.20057	1.000
After lift	743.52	.20143	.376
After transport	829.19	.20114	.379

C. Contact orientation changed late in development

Figure 6 compares each saved contact orientation with T7. For unit quaternions, the distance is

$$d_R(q_i, q_7) = 2 \arccos(\min(1, |q_i^\top q_7|)) . \quad (3)$$

The plot is a retrospective diagnostic: T7 is selected after observing success, and its zero is true by definition. Earlier attempts are approximately $8\text{--}10^\circ$ away; T6 is 2.39° away. This is consistent with the recorded orientation correction, but is not an isolated ablation of yaw or an object-relative error measurement. Door pose changes are not compensated.

D. Cap removal and held transport

The cap sequence in Fig. 3 produced 9.17 mm of measured vertical rise, followed by 107.42 mm of end-effector displacement. Opening remained near 0.201 (Table III). The normalized bottle-support shift after lift was 0.0095, and the source’s cap-like appearance dropped to 0.376 of its pre-lift value. Together with the images, these observations support removal and held transport. The distance is a robot end-effector measurement, not an independent high-precision cap tracker.

The observed cap was removed by a side pinch and lift. No threaded unscrewing, resealing, or verified placement is claimed. The local numerical precision above describes recorded values; it is not a calibrated accuracy statement for the camera or robot.

E. What the failures changed

Across the cases, progress involved changing what the system considered relevant evidence. The door moved from plausible closure to proof, then to checkpointed motion plus independent endpoint inspection. The cap moved from global appearance to an anchored object/support relation, then to separate removal and retention checks. Some changes were prompted directly by operator corrections. The result is therefore a human-guided code-development process with executable outcomes, not an unassisted discovery claim.

The saved full-pull record is incomplete: some aperture checks lived in memory and were lost when an exception interrupted result serialization. We retain phase summaries and reported slip stops rather than fabricating a continuous trace. Historical joint-torque warnings are sparse, absolute-valued samples; they cannot identify break-away force. These logging findings are part of the reproducibility audit, not additional physical trials.

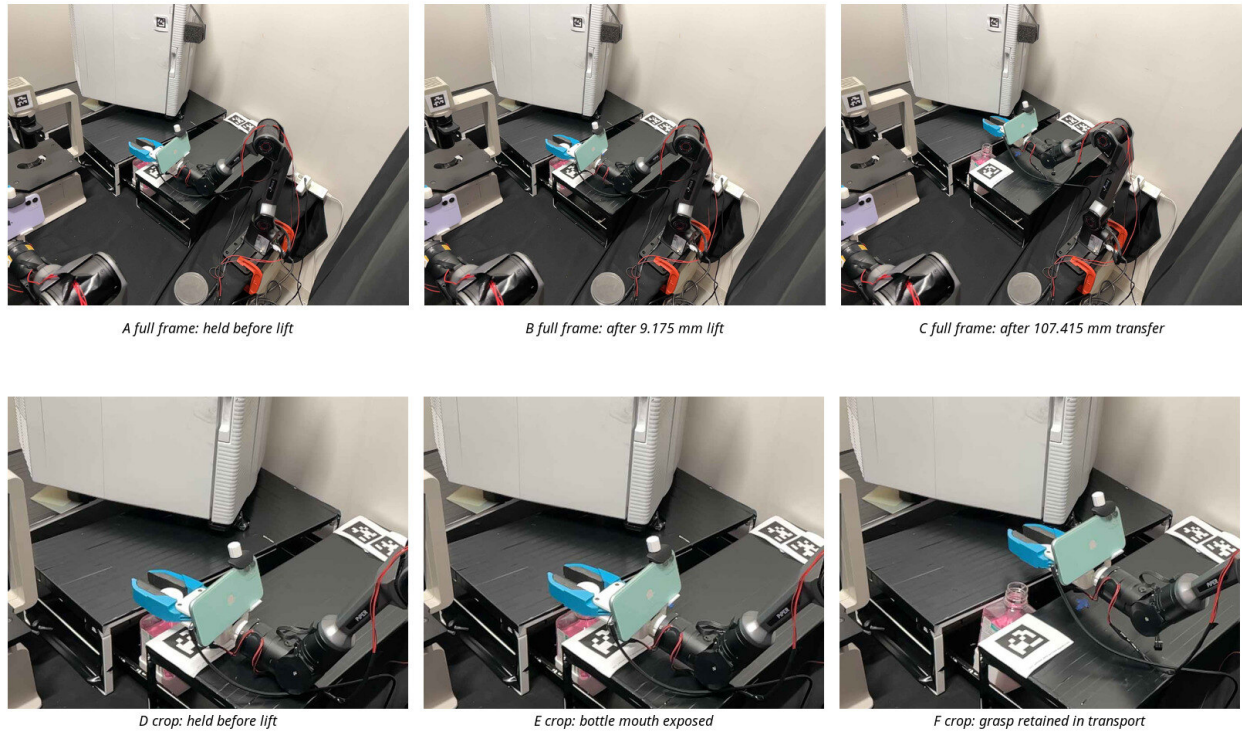


Fig. 3. One cap sequence, not three repetitions. Top: full fixed-head observations before lift, after lift, and after held transport. Bottom: corresponding detail crops. Measured end-effector rise was 9.17 mm; subsequent transport was 107.42 mm. Source appearance and bottle stability complement robot state, because end-effector motion alone does not prove cap removal.

VII. DISCUSSION AND LIMITATIONS

Why laboratory access matters. A manipulation primitive is useful to biological automation only if it establishes the required access state without confusing motion of the tool, target, and supporting container. Opening an incubator and exposing a bottle mouth are examples of this prerequisite. Sterility, contamination, sample integrity, and downstream biological outcomes were not evaluated. The work is a step toward laboratory automation, not a complete autonomous biological experiment.

What generalizes. Engagement/release/retention decomposition, contact-relative motion, and explicit support witnesses can be reused. The particular appearance models, calibrated frames, and validated cap route are local. Different handles, bottles, camera placements, or clutter require new evidence. No cross-laboratory transfer or broad reliability result is inferred from the two cases.

What remains to test. A prospective study should freeze an external endpoint evaluator and compare fixed controllers with code adaptation under matched resets, human-feedback budgets, and trial budgets. Continuous force or current logging, synchronized with door/cap motion, is needed to study break-away dynamics. Repeated trials with varied initial poses are needed to estimate reliability. The current results motivate those experiments without substituting chronological improvement for controlled causal evidence.

VIII. CONCLUSION

The important distinction in opening and removal is not simply whether the gripper moved or closed, but whether the intended physical constraint changed. A coding agent, guided by human corrections and physical evidence, converted archived door demonstrations and demonstration-free cap probes into executable controllers with task-specific verification. The resulting case studies show how robot learning can modify the representation of success as well as the motion used to reach it, while preserving an auditable boundary between a plausible action and a verified capability.

ACKNOWLEDGMENTS AND AI USE DISCLOSURE

OpenAI Codex assisted with manuscript drafting, analysis scripts, and figure layout throughout this article. Codex also generated and revised experimental robot code as the system under study. Experimental photographs are recorded camera images, not generative images. This disclosure distinguishes manuscript assistance from the robot-development intervention evaluated here.

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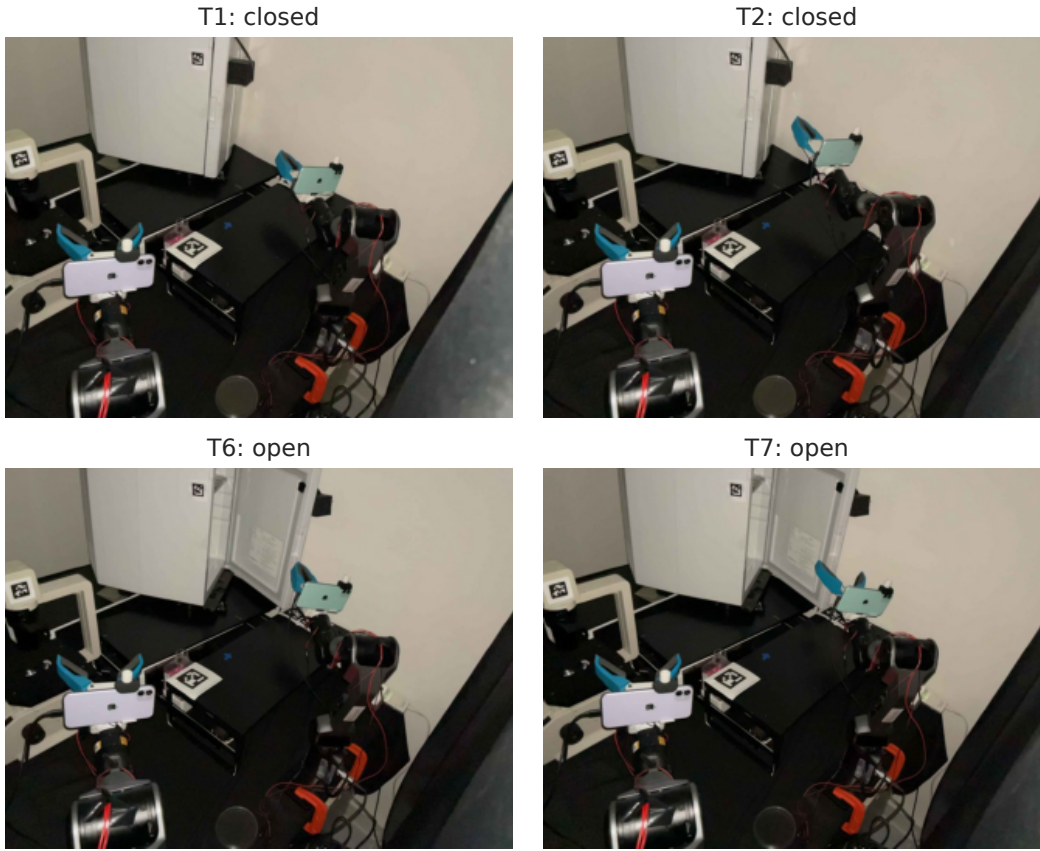


Fig. 4. Saved head views for the four compared door pulls. The upper row shows closed endpoints (T1/T2); the lower row shows open endpoints (T6/T7). These images are acquired after the pulls, not at matched intermediate times. The distinction remains visible even though all four later grasp observations indicate almost closed jaws.

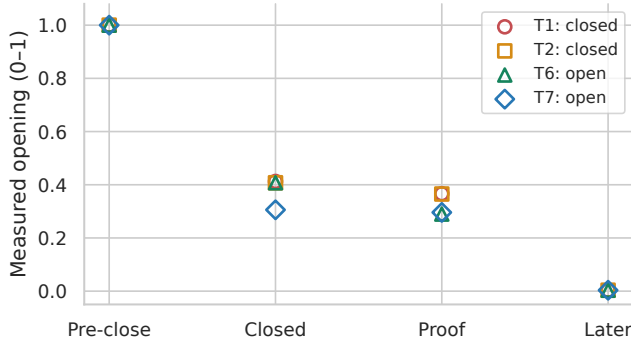


Fig. 5. The same final aperture accompanies different door outcomes. Each dot is a saved phase observation or summary; horizontal position is not time. No full-pull curve is reconstructed. Outcome labels use later images and depth, not aperture.

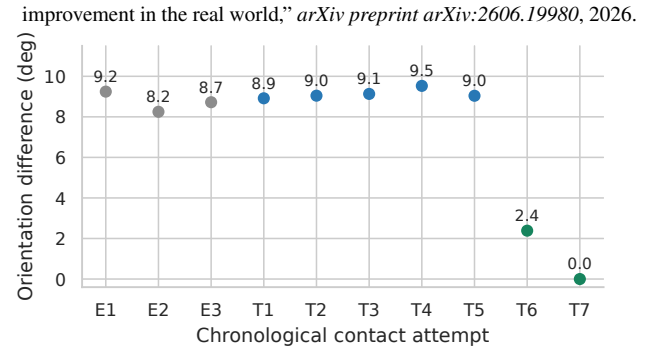


Fig. 6. Full contact-orientation difference to successful trial T7. Gray: early contact-only attempts; blue: T1–T5; green: T6/T7 open endpoints. T7 is a post hoc reference, not a target known to the earlier controller.

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